

Complex Exponential, Logarithm, and Powers

MA4410

Definition of $\exp(z)$

Definition

For $z \in \mathbb{C}$, the complex exponential is defined by the power series

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

This definition mirrors the real exponential and is taken as *the* definition in our text.

Theorem

The power series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

has infinite radius of convergence.

- Converges absolutely for all $z \in \mathbb{C}$.
- Defines an entire function.

Euler's Identity

For $z = i\theta$ with $\theta \in \mathbb{R}$,

$$\exp(i\theta) = \cos \theta + i \sin \theta.$$

In particular,

$$\exp(i\pi) + 1 = 0.$$

Theorem

For all $z, w \in \mathbb{C}$,

$$\exp(z + w) = \exp(z) \exp(w).$$

- Follows from absolute convergence of the defining power series.
- Fundamental to defining trigonometric and logarithmic functions in $\mathbb{C}$.

Definition of the Complex Logarithm

Let $z \neq 0$ with polar form $z = re^{i\theta}$.

Definition

The (multi-valued) complex logarithm is

$$\log z := \ln r + i(\theta + 2\pi k), \quad k \in \mathbb{Z}.$$

Definition

The principal logarithm is defined by restricting $\theta \in (-\pi, \pi)$:

$$\operatorname{Log} z := \ln r + i\theta.$$

This corresponds to removing the negative real axis as a branch cut.

Logarithmic Identities

For $z, w \neq 0$:

- $\operatorname{Log}(zw) = \operatorname{Log} z + \operatorname{Log} w (+ 2\pi ik)$
- $\operatorname{Log}(z^n) = n \operatorname{Log} z (+ 2\pi ik)$
- $\operatorname{Log}(zw) = \operatorname{Log} z + \operatorname{Log} w$ *may fail* if branch cuts are crossed

Definition of z^w

Definition (Definition 5.3)

For $z \neq 0$ and $w \in \mathbb{C}$,

$$z^w := \exp(w \log z).$$

Since $\log z$ is multi-valued, so is z^w in general.

Multiple Values: Examples

Example

$(-1)^{1/2}$:

$$\log(-1) = i(\pi + 2\pi k), \quad k \in \mathbb{Z}$$

$$(-1)^{1/2} = \exp\left(\frac{1}{2}i(\pi + 2\pi k)\right) = \pm i.$$

Another Example

Example

i^i :

$$\log i = i \left(\frac{\pi}{2} + 2\pi k \right)$$

$$i^i = \exp \left(-\frac{\pi}{2} - 2\pi k \right), \quad k \in \mathbb{Z}.$$

Infinitely many real-valued possibilities.

Check Your Intuition: Exponential

- If $z = x + iy$, what geometric effect does $\exp(z)$ have on the plane?
- How do $\exp(x)$ and $\exp(iy)$ separately affect magnitude and angle?
- Why can $\exp(z)$ never be zero?

Check Your Intuition: Logarithm

- Why must any logarithm in $\mathbb{C}$ be multi-valued?
- What goes wrong with $\log(zw) = \log z + \log w$ near the branch cut?
- How does going once around the origin change the value of $\log z$?

Check Your Intuition: Branches

- Why is the negative real axis a natural choice of branch cut for $\log z$?
- Could we choose a different ray as a branch cut? What would change?
- What must be true of a domain for a single-valued logarithm to exist?

Check Your Intuition: Complex Powers

- Why does z^w typically have infinitely many values?
- When does z^n (with $n \in \mathbb{Z}$) become single-valued?

Summary

- $\exp(z)$ is defined by a power series and is entire.
- $\log z$ is multi-valued and requires branch choices.
- Complex powers inherit multi-valued behavior from $\log z$